

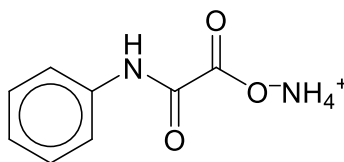
Eunoia Junior College
9729 H2 Chemistry
2018 Paper 3 Suggested Solution

- 1 (a) Describe and explain the trend in the thermal stabilities of the Group 2 carbonates. [3]

The thermal stability of the Group 2 carbonates increases down the Group from MgCO_3 to BaCO_3 .

Upon going down Group 2, the cationic radius increases, hence the charge density and polarising power of the M^{2+} cation decreases. As a result, the carbonate anionic electron cloud is less distorted and thus the C–O bond in the CO_3^{2-} anion is weakened to a lesser extent. Therefore the carbonate becomes more stable to thermal decomposition and the thermal stability increases.

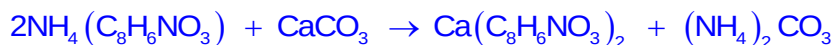
- (b) Buildings and statues constructed of limestone (calcium carbonate, CaCO_3) can be corroded by acid rain. They can be partially protected by painting them with a solution containing ammonium *N*-phenyloxamate.



ammonium *N*-phenyloxamate

The ammonium salt reacts with the surface of the limestone to product an inert layer of calcium *N*-phenyloxamate. The by-product, ammonium carbonate, is water soluble.

- (i) Write an equation for the reaction between calcium carbonate and ammonium *N*-phenyloxamate, $\text{NH}_4(\text{C}_8\text{H}_6\text{NO}_3)$. [1]



- (ii) The solubility product, K_{sp} , for calcium *N*-phenyloxamate is $1.75 \times 10^{-5} \text{ mol}^3 \text{ dm}^{-9}$.

Calculate the concentration of calcium *N*-phenyloxamate in a saturated solution in g dm^{-3} . [3]

Let the solubility of calcium *N*-phenyloxamate, $\text{Ca}(\text{C}_8\text{H}_6\text{NO}_3)_2$, be $x \text{ g dm}^{-3}$.

$$M(\text{Ca}(\text{C}_8\text{H}_6\text{NO}_3)_2) = 40.1 + 12.0 \times 16 + 1.0 \times 12 + 14.0 \times 2 + 16.0 \times 6 = 368.1 \text{ g mol}^{-1}$$

$$[\text{Ca}(\text{C}_8\text{H}_6\text{NO}_3)_2] = \frac{x}{368.1} \text{ mol dm}^{-3}$$

In a saturated solution,

$$K_{sp} = [\text{Ca}^{2+}][\text{C}_8\text{H}_6\text{NO}_3^-]^2 = \left(\frac{x}{368.1}\right)\left(2 \times \frac{x}{368.1}\right)^2$$

$$x^3 = \frac{K_{sp}}{4} \times 368.1^3 = \frac{1.75 \times 10^{-5}}{4} \times 368.1^3$$

$$x = 6.02$$

Concentration of calcium *N*-phenyloxamate in a saturated solution is **6.02 g dm⁻³**.

- (c) Suggest a synthesis of *N*-phenyloxamic acid starting from phenylamine and ethane-1,2-diol, according to the scheme in Fig. 1.1. Deduce the structure of compound **A**. Include reagents and conditions for all reactions, and the structures of all other intermediate compounds, in your answer.

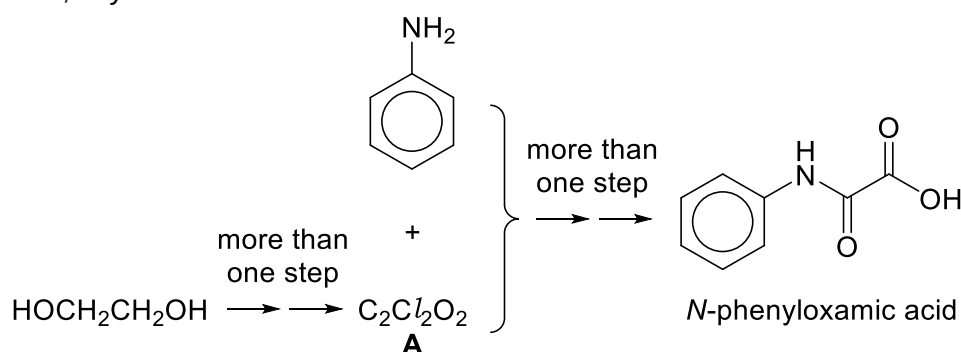
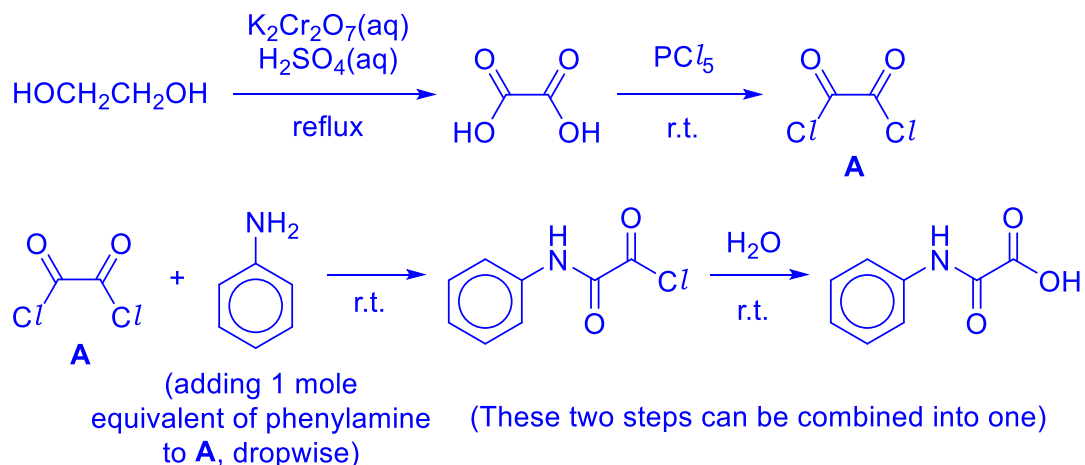


Fig. 1.1

[5]

A possible synthetic route is

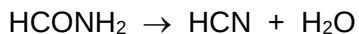


- (d) Explain why amides, RCONH_2 , are neutral, rather than basic.

[1]

Amides are neutral due to the effective delocalisation of the lone pair of electrons on the amide nitrogen into the C=O , rendering the lone pair not available for donation to an acid.

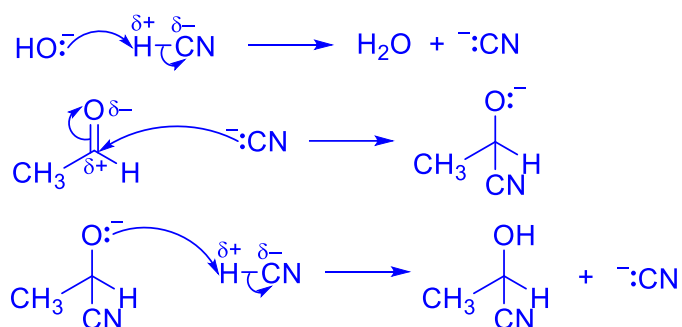
- (e) The simplest amide, methanamide, HCONH_2 , can be dehydrated to hydrogen cyanide, HCN .



- (i) Draw the mechanism of the reaction between hydrogen cyanide and ethanal, CH_3CHO , to form 2-hydroxypropanenitrile, $\text{CH}_3\text{CH}(\text{OH})\text{CN}$.

Show all charges and relevant lone pairs and show the movement of electron pairs by using curly arrows. [3]

Mechanism of the reaction:



- (ii) In some circumstances, double bonds will undergo a nucleophilic addition reaction.

Suggest reasons to explain Fig. 1.2.

Use the concepts of electronegativity, electronic and steric effects, and delocalisation in your answer.

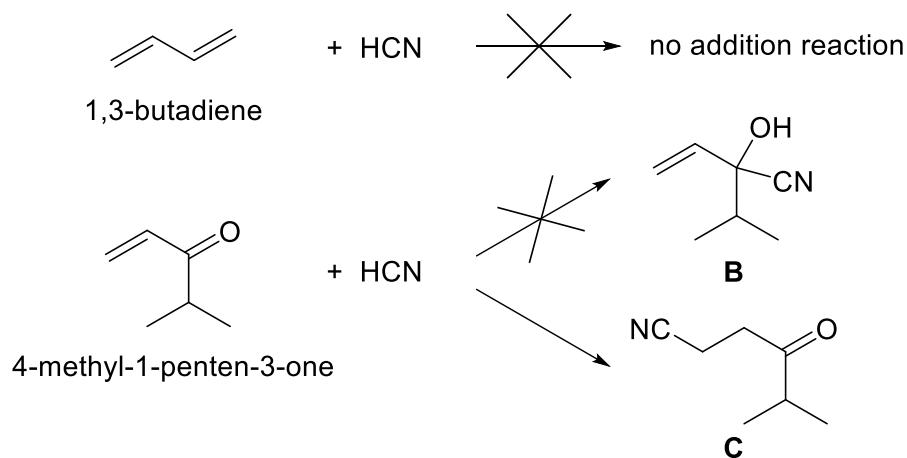


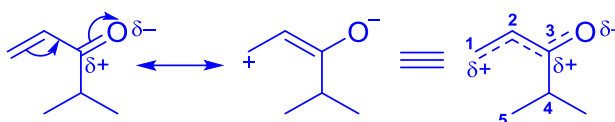
Fig. 1.2

[3]

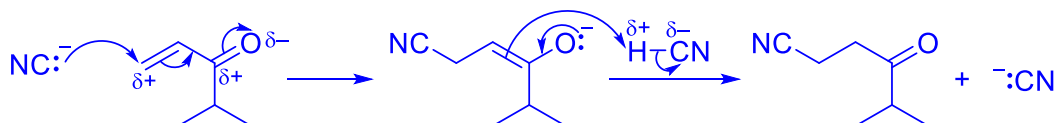
Since the C=C double bond is non-polar and electron-rich, it will repel an incoming CN^- nucleophile. In addition, HCN is a very weak acid, wherein the H–C bond is only very slightly polar due to the similar electronegativity of C and H, hence it does not add across the C=C. As a result, 1,3-butadiene does not react with HCN.

For 4-methyl-1-penten-3-one, the C=O double bond is polar as O is more electronegative compared to C, rendering the carbonyl carbon electron-deficient (electrophilic). However, due to steric hindrance from the bulky adjacent isopropyl ($-\text{CH}(\text{CH}_3)_2$) group, attack of the CN^- nucleophile at the carbonyl carbon is hindered and HCN is unable to add across the C=O to give **B**, like in ethanal.

On the other hand, the π electrons of the C=C in 4-methyl-1-penten-3-one can be delocalised into the conjugated polar $\text{C}=\text{O}$:

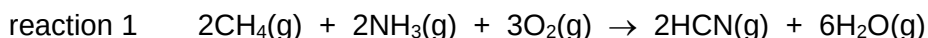


resulting in C-1 attaining a partial positive charge. C-1 being at the terminal is readily accessible by the CN^- nucleophile, resulting in addition of HCN to give **C**:



[Total: 19]

- 2 (a) Hydrogen cyanide, HCN, is made industrially by oxidising a mixture of methane and ammonia over a platinum catalyst at 1000°C .



During this reaction, the oxidation number of hydrogen remains at +1.

- (i) State the oxidation numbers of carbon, nitrogen and oxygen in the reactants and in the products in reaction 1. [2]

The oxidation numbers are:

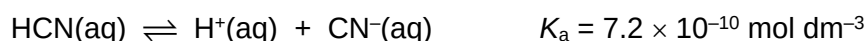
element	reactants	products
C	-4	+2
N	-3	-3
O	0	-2

- (ii) Calculate the enthalpy change, ΔH_r , for reaction 1, using bond energies from the *Data Booklet*. [2]

The enthalpy change for reaction 1 is

$$\begin{aligned}\Delta H_r &= \sum \text{B.E. (bonds broken)} - \sum \text{B.E. (bonds formed)} \\ &= 8\text{B.E. (C-H)} + 6\text{B.E. (N-H)} + 3\text{B.E. (O=O)} \\ &\quad - [2\text{B.E. (C-H)} + 2\text{B.E. (C}\equiv\text{N)} + 12\text{B.E. (O-H)}] \\ &= 8 \times 410 + 6 \times 390 + 3 \times 496 - (2 \times 410 + 2 \times 890 + 12 \times 460) \\ &= 7108 - 8120 \text{ kJ mol}^{-1} \\ &= -1012 \text{ kJ mol}^{-1}\end{aligned}$$

HCN is a weak acid.



A mixture containing both HCN and CN^- ions is a useful reagent in organic chemistry.

- (iii) Calculate the $[\text{CN}^-]/[\text{HCN}]$ ratio in such a mixture whose pH is 10.0. [2]

Using the K_a , at pH 10.0,

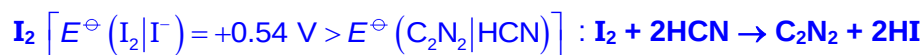
$$\begin{aligned}K_a &= \frac{[\text{H}^+][\text{CN}^-]}{[\text{HCN}]} \\ 7.2 \times 10^{-10} &= [\text{H}^+] \times \frac{[\text{CN}^-]}{[\text{HCN}]} = 10^{-10.0} \times \frac{[\text{CN}^-]}{[\text{HCN}]} \\ \frac{[\text{CN}^-]}{[\text{HCN}]} &= \underline{\underline{7.20}}\end{aligned}$$

- (b) HCN can be oxidised to cyanogen, C_2N_2 .



- (i) Suggest a suitable oxidant to oxidise HCN to C_2N_2 in acidic solution, using the *Data Booklet*.

Write an equation for the reaction and calculate the $E_{\text{cell}}^\ominus$. [2]



$$\begin{aligned}E_{\text{cell}}^\ominus &= E^\ominus (\text{I}_2|\text{I}^-) - E^\ominus (\text{C}_2\text{N}_2|\text{HCN}) \\ &= +0.54 - (+0.37) = \underline{\underline{+0.17 \text{ V}}}\end{aligned}$$

- (ii) Draw a dot-and-cross diagram of the cyanogen molecule, and state the hybridisation of the carbon atoms in it. [2]

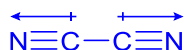


The carbon atom in cyanogen is sp hybridised.

- (iii) Nitrogen is more electronegative than carbon.
Explain how this affects the polarity of the bonds in the cyanogen molecule, and deduce the overall polarity of the molecule. [2]

The $\text{C}\equiv\text{N}$ will be a polar bond with the dipole pointing in the direction of N, since N is more electronegative than C, while the C–C bond will be non-polar.

Due to the hybridisation of the two C atoms, cyanogen will be a linear molecule, wherein the two dipoles due to the $\text{C}\equiv\text{N}$ bond will point in opposite direction and hence cancels off one another, resulting in cyanogen being a non-polar molecule:



- (c) Cyanogen chloride, ClCN , reacts with an excess of water in the following manner.



The reaction is faster if hydroxide ions are present.



Both of these are *bimolecular* reactions: in each case their rate-determining step involves two species (molecules or ions).

The rate constants for these two reactions have been measured.

for reaction 2, $k_2 = 5.1 \times 10^{-7} \text{ s}^{-1}$

for reaction 3, $k_3 = 4.2 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$

- (i) Name the *type of reaction* that is occurring during these two reactions. [1]

Nucleophilic substitution

- (ii) Suggest why the units of k_2 are s^{-1} , despite its rate-determining step being a bimolecular reaction. [1]

This is because the water is in excess, hence the concentration of water does not change significantly for the reaction. The reaction thus becomes pseudo-zero order with respect to H_2O , and pseudo-first order overall. Therefore the units of k_2 is s^{-1} .

- (iii) Deduce whether H_2O is acting as an electrophile or nucleophile in reaction 2. Explain your answer. [1]

H_2O is acting as a **nucleophile** because reaction 2 involves the H_2O molecule **donating a lone pair of electrons** to the carbon in $\text{C}\text{I}\text{CN}$, displacing the chlorine as a Cl^- leaving group, which then combines with a H^+ to give HCl .

- (iv) Suggest the rate equation for reaction 3. [1]

$$\text{rate} = k_3 [\text{C}\text{I}\text{CN}] [\text{OH}^-]$$

When 0.010 mol of $\text{C}\text{I}\text{CN}$ is dissolved in 100 cm^3 of a solution buffered at pH 10.0, both reactions 2 and 3 take place.

- (v) Calculate the ratio of the initial rates of reactions 2 and 3 under these conditions. [3]

rate equation for reaction 2 is $\text{rate} = k_2 [\text{C}\text{I}\text{CN}]$

$$\text{initial } [\text{C}\text{I}\text{CN}] = \frac{0.010 \text{ mol}}{\frac{100}{1000} \text{ dm}^3} = 0.100 \text{ mol dm}^{-3}$$

$$\text{initial rate of reaction 2} = k_2 [\text{C}\text{I}\text{CN}] = 5.1 \times 10^{-7} \times 0.100 = 5.10 \times 10^{-8} \text{ mol dm}^{-3} \text{ s}^{-1}$$

$$\begin{aligned} \text{initial rate of reaction 3} &= k_3 [\text{C}\text{I}\text{CN}] [\text{OH}^-] = 4.2 \times 0.100 \times 10^{-(14.0-10.0)} \\ &= 4.20 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1} \end{aligned}$$

$$\text{ratio of initial rates of reactions 2 and 3} = 5.10 \times 10^{-8} : 4.20 \times 10^{-5} = \underline{\underline{1:824}}$$

- (d) The products of reaction 2 are acidic, whereas the reactants are neutral.

Outline a suitable experiment for determining a value for k_2 . [3]

Using a data-logger with a pH probe, track the change in pH of the reaction mixture with time, upon addition of a known amount of $\text{C}\text{I}\text{CN}$ to a known volume of deionised water in a beaker with magnetic stirring, until the pH plateaus to a constant value.

Convert pH to $[\text{H}^+]$ using the relationship $[\text{H}^+] = 10^{-\text{pH}}$, and plot $[\text{H}^+]$ against time.

Either determine the initial rate of the reaction by determining the gradient of the tangent to the curve at $t = 0$, the time of addition of $\text{C}\text{I}\text{CN}$ to water.

Determine the initial concentration of $\text{C}\text{I}\text{CN}$ using the amount of $\text{C}\text{I}\text{CN}$ and water used.

Then k_2 is given by $\frac{\text{initial rate}}{\text{initial } [\text{C}\text{I}\text{CN}]}$.

Alternatively, determine the half-life, $t_{\frac{1}{2}}$, for the reaction from the plot of $[H^+]$ against time.

Then k_2 is given by $\frac{\ln 2}{t_{\frac{1}{2}}}$.

- (e) Cyanogen bromide, BrCN, undergoes addition reactions with alkenes. With propene, isomer **D** is produced rather than isomer **E**.

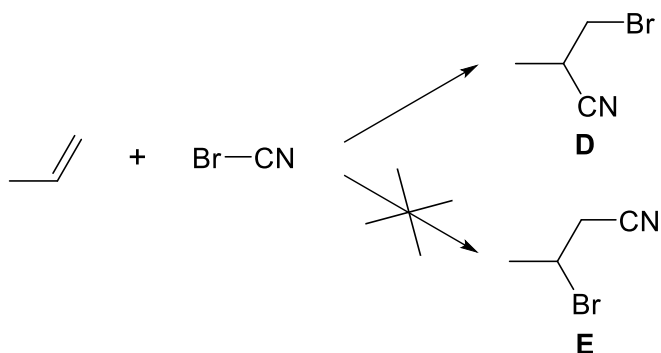
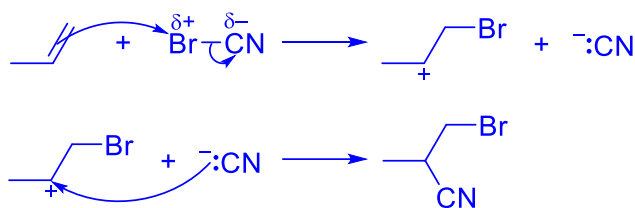


Fig. 2.1

Suggest a mechanism for this reaction and use it to explain the preferential production of isomer **D**. [3]

The mechanism is



The BrCN molecule is polarised in a way such that the bromine carries the partial positive charge due to the higher electronegativity of N compared to Br. Initial nucleophilic attack by the π electrons of the C=C will be at this electron-deficient Br atom, leading to formation of a carbocation intermediate with a C-Br bond, and the CN⁻ nucleophile.

The Br attaches itself to C-1 rather than C-2 as this gives rise to a more stable 2° carbocation intermediate, $\text{CH}_3\text{-C}^+(\text{Br})\text{-CH}_3$, rather than the 1° carbocation, $\text{CH}_3\text{-CH}^+\text{-CH}_2\text{-Br}$, if Br is to attach itself to C-2.

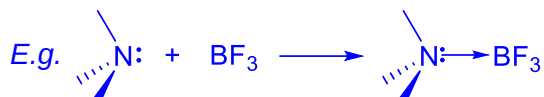
Subsequent attack of the CN⁻ nucleophile at the carbocationic centre gives isomer **D**. The less stable 1° carbocation will give isomer **E**, which is not observed.

[Total: 25]

3 (a) Amines are Lewis bases.

- (i) Explain what is meant by this statement. Illustrate your answer with an equation for a suitable reaction of an amine of your choice. [2]

The statement means that amines possess lone pair(s) of electrons to can be donated to an electron-deficient centre.



- (ii) Describe and explain the relative basicities of methylamine, dimethylamine, and trimethylamine in the gas phase. [3]

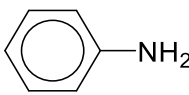
In the gas phase, the relative basicities is as follows:

methylamine < dimethylamine < trimethylamine

This is because the methyl group is an electron-donating group. With increasing number of methyl groups attached to the nitrogen, the nitrogen atom becomes increasingly more electron-rich and hence the lone pair of electrons on the nitrogen become more available for donation to an acid, accounting for the increase in gas phase basicity.

(b) The K_b values of four bases, at 25 °C, are shown in Table 3.1.

Table 3.1

base	formula	$K_b/\text{mol dm}^{-3}$
ammonia	NH_3	1.8×10^{-5}
ethylamine	$\text{CH}_3\text{CH}_2\text{NH}_2$	4.5×10^{-4}
phenylamine		7.4×10^{-10}
4-chlorophenylamine		9.6×10^{-11}

- (i) Write the K_b expression for ethylamine. [1]

$$K_b = \frac{[\text{CH}_3\text{CH}_2\text{NH}_3^+][\text{OH}^-]}{[\text{CH}_3\text{CH}_2\text{NH}_2]}$$

- (ii) Explain the relative magnitudes of the K_b values in Table 3.1. [3]

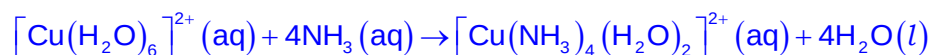
Ethyl group is an electron-donating group, which will increase the electron-density on the nitrogen atom, making the lone pair of electrons on nitrogen more available for donation to the hydrogen of water, thus a higher K_b compared to ammonia.

On the other hand, the lone pair of electrons on nitrogen can be delocalised into the benzene ring in the case of phenylamine and 4-chlorophenylamine. Thus makes the lone pair of electrons less available for donation to the hydrogen of water, explaining for their lower K_b compared to ammonia.

The additional electron-withdrawing Cl atom on the benzene ring of 4-chlorophenylamine causes the lone pair of electrons on the nitrogen to be more readily delocalised into the benzene ring, hence even less available for donation to the hydrogen of water compared to phenylamine, hence the least basic, with the lowest K_b .

(c) The addition of $\text{NH}_3(\text{aq})$ to a solution containing $\text{Cu}^{2+}(\text{aq})$ produces a deep blue solution.

(i) Write an equation for this reaction, and state the *type of reaction* occurring. [2]



This is a ligand exchange reaction.

This solution changes from deep blue to purple when ethane-1,2-diamine, $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$, is added.

(ii) Suggest an explanation for this observation and write an equation for the reaction occurring.

[The formula of ethane-1,2-diamine can be shortened to **en** in the formulae and equations of your answer.] [2]

As en is a bidentate ligand, only three en molecules is required to bind to each Cu^{2+} ion. As a result, there is an increase in the number of aqueous species since six monodentate ligands are displaced. Thus entropically, ligand exchange is favourable, leading to formation of the purple $[\text{Cu}(\text{en})_3]^{2+}$ complex:



(d) The melting points of pentane, butylamine, propanoic acid and glycine increase, as shown in Table 3.2.

Table 3.2

compound	formula	M_r	melting point/ $^{\circ}\text{C}$
pentane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	72	-130

butylamine	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$	73	-49
propanoic acid	$\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$	74	-21
glycine	$\text{H}_2\text{NCH}_2\text{CO}_2\text{H}$	75	290

Suggest explanations for these differences.

[3]

Pentane is a non-polar hydrocarbon, in which only weak instantaneous dipole-induced dipole attractions occur between pentane molecules, hence only a small amount of energy is needed to overcome the weak dispersion force holding the molecules in place in solid pentane, thus the low melting point.

For butylamine, propanoic acid and glycine, they have similar molecular mass as pentane and hence similar number of electrons in each molecule, thus the dispersion forces between the molecules for all four compounds will be comparable.

On top of that, butylamine is able to form hydrogen bonds with each other since it possesses highly polar N-H bonds, wherein the hydrogen can form hydrogen bonds with the lone pair of electrons on the nitrogen of a neighbouring molecule. Such hydrogen bonds are much stronger than the dispersion forces and hence more energy is required to overcome the attractive intermolecular forces between the butylamine molecules, resulting in a significantly higher melting point compared to pentane.

Similar to butylamine, propanoic acid is able to form hydrogen bonds involving the hydrogen atom of the O-H bond with the lone pair of electrons on either of the CO_2H oxygen atoms. Due to the slightly higher electronegativity of O compared to N, the O-H is more polar and hence the hydrogen bonds formed by the O-H hydrogen will be slightly stronger than that of the N-H hydrogen. Slightly more energy is needed to overcome this stronger hydrogen bond and hence the melting point of propanoic acid is slightly higher than that of butylamine.

Lastly, glycine being an α -amino acid exists in the zwitterionic form in the solid state, with strong electrostatic attractions between the $-\text{CO}_2^-$ and $-\text{NH}_3^+$ groups between two glycine molecules. As a result, large amount of energy is needed to overcome this much stronger ionic interactions between molecules, thus accounting for the much higher melting point observed.

[Total: 16]

- 4 (a) (i) State how the reactivity of the halogens as oxidising agents varies down the group, and relate this variation to relevant $E^\ominus$ values. [2]

Down the group, the halogens become weaker oxidising agent, which can be seen from the less positive $E^\ominus(\text{X}_2|\text{X}^-)$, from +1.36V ($\text{X} = \text{Cl}$) to +1.07V ($\text{X} = \text{Br}$) to +0.54V ($\text{X} = \text{I}$). An oxidising agent itself is reduced in the process of oxidising another species. A less positive $E^\ominus$ means that reduction of the halogen to the halide is less favourable, and thus lower oxidising strength.

- (ii) Describe a reaction that illustrates the relative oxidising abilities of two halogens of your choice. [1]

Consider the reaction between Cl_2 and I_2 with the thiosulfate(VI) ion, $\text{S}_2\text{O}_3^{2-}$.

Cl_2 is capable of oxidising the $\text{S}_2\text{O}_3^{2-}$ to SO_4^{2-} , where the oxidation state of S increases from +2 to +6.

On the other hand, I_2 is only capable of oxidising $\text{S}_2\text{O}_3^{2-}$ to the tetrathionate, $\text{S}_4\text{O}_6^{2-}$ ion, where the oxidation state of S increases from +2 to +2.5.

In both cases, the halogen X_2 is reduced to the corresponding halide ion, X^- , hence illustrating the higher oxidising ability of Cl_2 compared to I_2 .

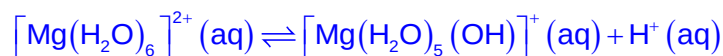
- (b) Magnesium, aluminium and silicon are three adjacent elements in the Periodic Table.

- (i) Describe what you would observe when water is added to separate samples of the chlorides of these three elements. Suggest the pH of the resulting solutions, and write equations where appropriate. [3]

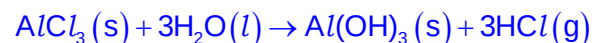
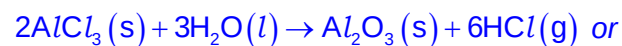
MgCl_2 will dissolve in the water added to give a colourless solution:



The solution is very slightly acidic at pH 6.5 due to slight hydrolysis of $\text{Mg}^{2+}(\text{aq})$ ion:

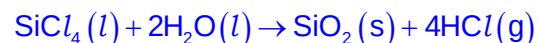


Anhydrous AlCl_3 reacts violently to give white fumes of HCl when water is added, with the formation of a white Al_2O_3 / $\text{Al}(\text{OH})_3$:



The solution will be acidic at pH 3 due to the HCl produced.

SiCl_4 reacts violently to give white fumes of HCl when water is added, with the formation of white SiO_2 :



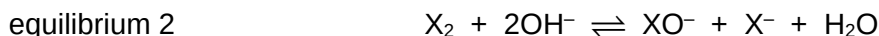
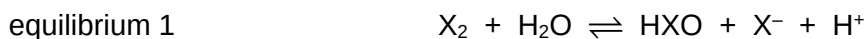
The solution will be acidic at pH 2 due to the HCl produced.

- (ii) Explain the differences in the reactions of water with magnesium chloride and with silicon chloride in terms of the structures of the chlorides. [1]

Magnesium chloride has a **giant ionic structure** consisting of Mg^{2+} and Cl^{-} ions, which undergoes **hydration** to give the respective hydrated ions. On the other hand, silicon tetrachloride has a **simple covalent structure** consisting of tetrahedral SiCl_4

molecules with polar $\overset{\delta+}{\text{Si}}-\overset{\delta-}{\text{Cl}}$ bonds, which undergoes **hydrolysis** with water, leading to formation of Si-OH bond and HCl.

- (c) Chlorine, bromine and iodine undergo reactions both with water, and with sodium hydroxide.



- (i) Write the expression for the equilibrium constant, K_c , for equilibrium 1, stating its units. [1]

$$K_c = \frac{[\text{HXO}][\text{X}^-][\text{H}^+]}{[\text{X}_2]} \text{ mol}^2 \text{ dm}^{-6}$$

A 0.10 mol sample of iodine, I_2 , was stirred with 1.0 dm³ of 0.50 mol dm⁻³ NaOH until equilibrium had been reached. The final concentration of IO^- was found to be 9.7×10^{-2} mol dm⁻³.



- (ii) Use these data to calculate a value for the equilibrium constant for this reaction. [4]

$$\text{initial } n_{\text{OH}^-} = 0.50 \text{ mol dm}^{-3} \times 1.0 \text{ dm}^3 = 0.500 \text{ mol}$$

$$\begin{aligned} n_{\text{I}^-} \text{ at equilibrium} &= n_{\text{I}_2} \text{ reacted} = n_{\text{IO}^-} \text{ at equilibrium} \\ &= 9.7 \times 10^{-2} \text{ mol dm}^{-3} \times 1.0 \text{ dm}^3 = 9.70 \times 10^{-2} \text{ mol} \end{aligned}$$

$$n_{\text{I}_2} \text{ at equilibrium} = 0.10 - 9.70 \times 10^{-2} = 3.00 \times 10^{-3} \text{ mol}$$

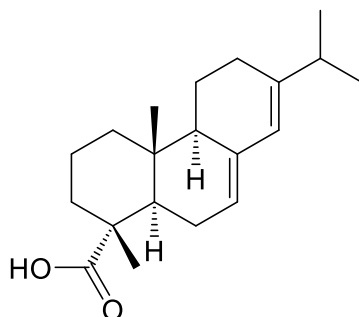
$$n_{\text{OH}^-} \text{ at equilibrium} = 0.50 - 2(9.70 \times 10^{-2}) = 0.306 \text{ mol}$$

At equilibrium,

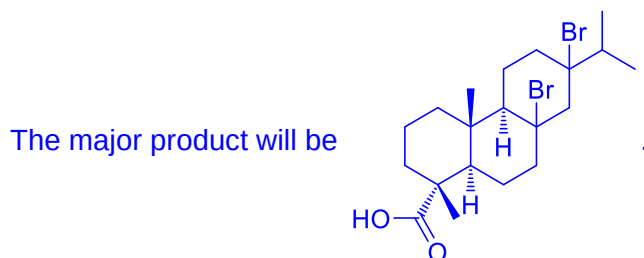
$$[\text{I}^-] = [\text{IO}^-] = 9.7 \times 10^{-2} \text{ mol dm}^{-3}; [\text{I}_2] = 3.00 \times 10^{-3} \text{ mol dm}^{-3}; [\text{OH}^-] = 0.306 \text{ mol dm}^{-3}$$

$$K_c = \frac{[\text{IO}^-][\text{I}^-]}{[\text{I}_2][\text{OH}^-]^2} = \frac{(9.7 \times 10^{-2})^2}{(3.00 \times 10^{-3})(0.306)^2} = \underline{\underline{33.5 \text{ mol}^{-1} \text{ dm}^3}}.$$

- (d) Salts of the oxyanions of chlorine are used as oxidants in fireworks. In a typical fireworks. In a typical firework, potassium chlorate(VII), KClO_4 , is used to oxidise an organic resin. Abietic acid is a major component of many resins.

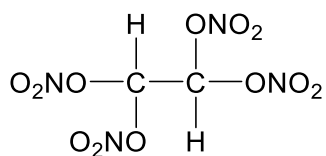
abietic acid, $C_{20}H_{30}O_2$

- (i) Draw the structure of the major product when abietic acid is reacted with an excess of HBr. State how many **extra** chiral centres are formed. [2]



2 extra chiral centres are formed.

When abietic acid reacts with potassium chlorate(VII) the products are carbon dioxide, steam and potassium chloride. In some situations, the non-volatile potassium chloride produced during this oxidation is a problem. This can be overcome by using an organic oxidant instead. It has been suggested that the recently-discovered solid compound tetranitrateoethane could be used.



tetranitrateoethane

- (ii) Explain why KCl is not volatile. [2]

KCl is not volatile since it has a giant ionic structure with K^+ and Cl^- ions held by strong electrostatic forces of attraction, thus large amount of energy is needed to overcome the strong ionic bonds to vapourise KCl .

- (iii) Calculate the percentage by mass of oxygen in tetranitrateoethane. [1]

Molecular formula is $C_2H_2N_4O_{12}$.

$$\text{Percentage by mass of O} = \frac{16.0 \times 12}{12.0 \times 2 + 1.0 \times 2 + 14.0 \times 4 + 16.0 \times 12} = \frac{192.0}{274.0} = \underline{70.1\%}$$

- (iv) Write an equation for the exothermic oxidation of powdered charcoal (carbon) by tetranitrateoethane. The products are carbon dioxide, steam and nitrogen gas. [1]



The entropy change for the reaction between powdered charcoal (carbon) and tetranitrateoethane is large and positive.

- (v) Suggest **two** reasons why the entropy change for the reaction is large and positive. Explain your answer.

The entropy change for the reaction is large and positive as the reaction leads to a large increase in the amount of gaseous products from solid reactants, hence there is a greater number of ways the energy of the system can be distributed. Secondly, due to the exothermic nature of the reaction, the entropy of the resultant hot gaseous products are expected to be large due to the increase in thermal energy of the particles which also increases the number of ways the energy can be distributed.

[Total: 20]

- 5 (a) (i) Explain why transition metal complexes are often coloured. [3]

In the presence of the coordinated ligands in transition metal complexes, the 3d orbitals of the central transition metal ion will be split into two sets of different energies (d-orbital splitting). In most transition metal ions, except for those with a d^0 or d^{10} configuration, the 3d orbitals will be partially filled. Electron(s) in a lower energy 3d orbital can absorb light energy, corresponding to wavelength in the visible region of the electromagnetic spectrum, and are promoted to the higher energy 3d orbital (d-d transition). As a result, the transmitted light will show the complementary colour of the colour absorbed, resulting in transition metal complexes being often coloured.

An acidic solution of vanadium(III) sulfate, $\text{V}_2(\text{SO}_4)_3$, is coloured, whereas a solution of sodium vanadate(V), NaVO_3 , is colourless.

- (ii) Explain these observations by considering relevant electronic configurations. [2]

Vanadium in the +3 oxidation state has a $[\text{Ar}] 3d^3$ configuration, while vanadium in the +5 oxidation state has a $[\text{Ar}] 3d^0$ configuration. A solution of $\text{V}_2(\text{SO}_4)_3$ contained the $\text{V}^{3+}(\text{aq})$ ion, where the V^{3+} ion is coordinated to 6 water ligands to give $[\text{V}(\text{H}_2\text{O})_6]^{3+}$. Due to the presence of a partially filled 3d subshell, d-d transition described in (a)(i)

is possible, leading to the solution being coloured. On the other hand, VO_3^- will be colourless as there are no 3d electrons, hence no d-d transition is possible. Electronic transitions corresponding to promotion of 3p electrons to the 3d orbitals will be much higher in energy, within the ultraviolet region of the electromagnetic spectrum.

- (b) When a solution of sodium vanadate(V) is gradually acidified, the colour changes from colourless to orange to red, and finally to a yellow solution which contains the ion VO_2^+ .

The standard reduction potentials, $E^\ominus$, for the electrode reactions $\text{VO}_3^-/\text{VO}^{2+}$ and $\text{VO}_2^+/\text{VO}^{2+}$ are both +1.00 V.

- (i) Suggest how the oxidising abilities of VO_3^- and VO_2^+ will change as pH increases, and hence suggest which will be the more powerful oxidant at pH 7. Explain your answer. [2]

The half-equation for reduction of VO_3^- to VO^{2+} is



while the half-equation for reduction of VO_2^+ to VO^{2+} is



As pH increases, $[\text{H}^+]$ decreases. Since (1) involves 4 moles of H^+ as reactant per mole of VO_3^- , while (2) involves only 2 moles of H^+ , equilibrium (1) will be more affected by the decrease in $[\text{H}^+]$, causing the position of equilibrium (1) to **shift more to the left** compared to equilibrium (2) when $[\text{H}^+]$ decreases.

Hence, the ease of reduction of VO_3^- to VO^{2+} decreases more than that of VO_2^+ , resulting in $E(\text{VO}_3^-|\text{VO}^{2+}) < E(\text{VO}_2^+|\text{VO}^{2+}) < +1.00\text{V}$, at pH above 0.

Therefore at pH 7, where $[\text{H}^+] = 10^{-7} < 1.00 \text{ mol dm}^{-3}$, VO_2^+ will be a more powerful oxidant.

- (ii) Predict the products of, and write an equation for the reaction that occurs when $\text{SO}_2(\text{aq})$ is added to an acidified solution containing $\text{VO}_3^-(\text{aq})$ ions. [2]

$$E^\ominus(\text{VO}_3^-|\text{VO}^{2+}) = +1.00\text{V}$$

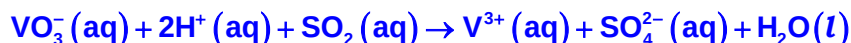
$$E^\ominus(\text{V}^{3+}|\text{V}^{2+}) = -0.26\text{V}$$

$$E^\ominus(\text{SO}_4^{2-}|\text{SO}_2) = +0.17\text{V}$$

Since $E^\ominus(\text{SO}_4^{2-}|\text{SO}_2)$ is less positive than $E^\ominus(\text{VO}_3^-|\text{VO}^{2+})$ and $E^\ominus(\text{VO}^{2+}|\text{V}^{3+})$, SO_2 will be able to reduce VO_3^- to VO^{2+} , and VO^{2+} to V^{3+} , while itself is oxidised to SO_4^{2-} , as the $E_{\text{cell}}^\ominus$ will be positive.

However, SO_2 will not be able to reduce V^{3+} to V^{2+} since $E^\ominus(\text{SO}_4^{2-}|\text{SO}_2)$ is more positive than $E^\ominus(\text{V}^{3+}|\text{V}^{2+})$, so $E_{\text{cell}}^\ominus$ will be negative.

The overall equation is:

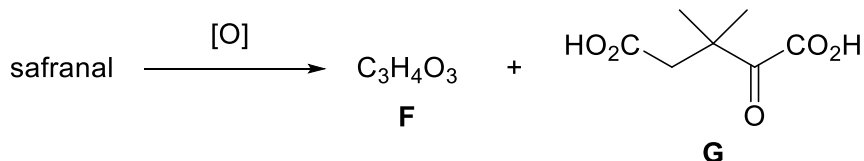


- (c) Saffron is a yellow spice isolated from the plant *crocus sativus*. It is used as a food colouring and flavouring agent.

The compound mainly responsible for the smell of saffron is safranal, $\text{C}_{10}\text{H}_{14}\text{O}$.

The molecule of safranal contains a 6-membered ring.

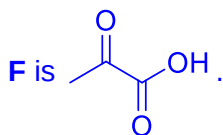
When safranal is heated with acidified manganate(VII) ions, two compounds **F** and **G** are formed.



Compound **F** gives a yellow precipitate with alkaline aqueous iodine.

- (i) Suggest the structure of compound **F**.

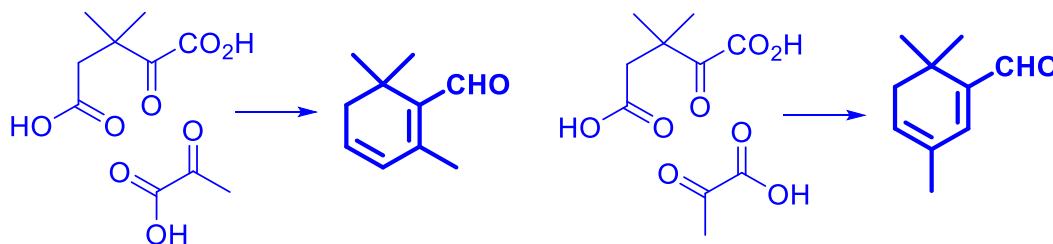
[1]



- (ii) Suggest **two** structures of safranal that are consistent with the information provided.

[2]

The two possible structures of safranal are



- (iii) Suggest the molecular formula of the product formed when safranal is reacted with an excess of hydrogen gas over a nickel catalyst. [1]

Safranal, $C_{10}H_{14}O$, has 3 double bonds (one $C=O$ and 2 $C=C$), thus $3H_2$ molecules can be added to give saturated $C_{10}H_{20}O$.

- (d) Compound **G** can be made from 3-bromo-2,2-dimethylpropanal by a four-step synthesis, via a key intermediate **J**.

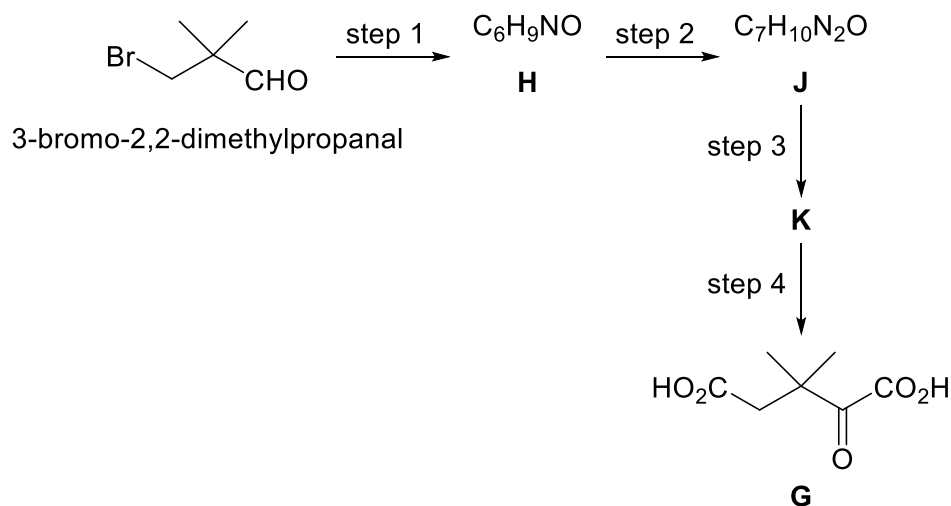


Fig. 5.1

Compound **J** is neutral and does not react with 2,4-dinitrophenylhydrazine or bromine water. It does, however, react with sodium metal.

- (i) Suggest what these observations indicate about the functional groups in **J**. [1]

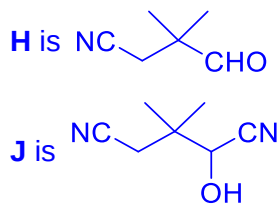
J is neutral, yet containing N, means it should be an amide or a nitrile.

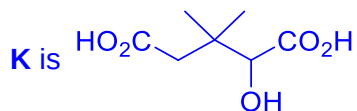
J does not react with 2,4-dinitrophenylhydrazine means that the carbonyl group (aldehyde and ketone) is absent.

J does not react with bromine water indicates that $C=C$, phenol and phenylamine is absent.

J reacts with sodium metal means that alcohol (or phenol) or carboxylic acid is present.

- (ii) Suggest structures for compounds **H**, **J** and **K**. [3]





(iii) Suggest reagents and conditions for steps 1, 3 and 4 in Fig. 5.1.

[3]

Step 1

Reagent: **ethanolic NaCN**

Condition: **heat**

Step 3

Reagent: **dilute sulfuric acid**

Condition: **heat with reflux**

Step 4

Reagent: **aqueous K₂Cr₂O₇, aqueous H₂SO₄**

Condition: **heat with reflux**

[Total: 20]